

Differentiable Karplus-Strong

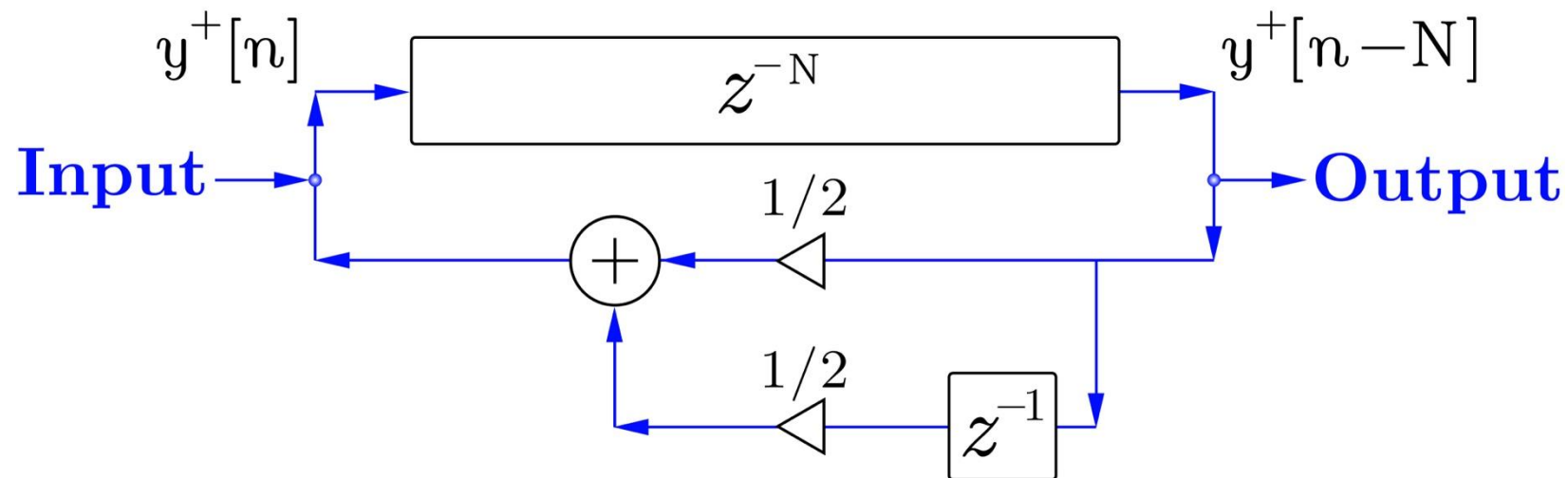
@DMRN+20

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Karplus-Strong Algorithm

- An algorithm for plucked-string synthesis.
- Input is usually a burst of noise of N samples.
- Fundamental Frequency (f_0) is controlled through delay length $N = \frac{f_s}{f_0} - \frac{1}{2}$ samples



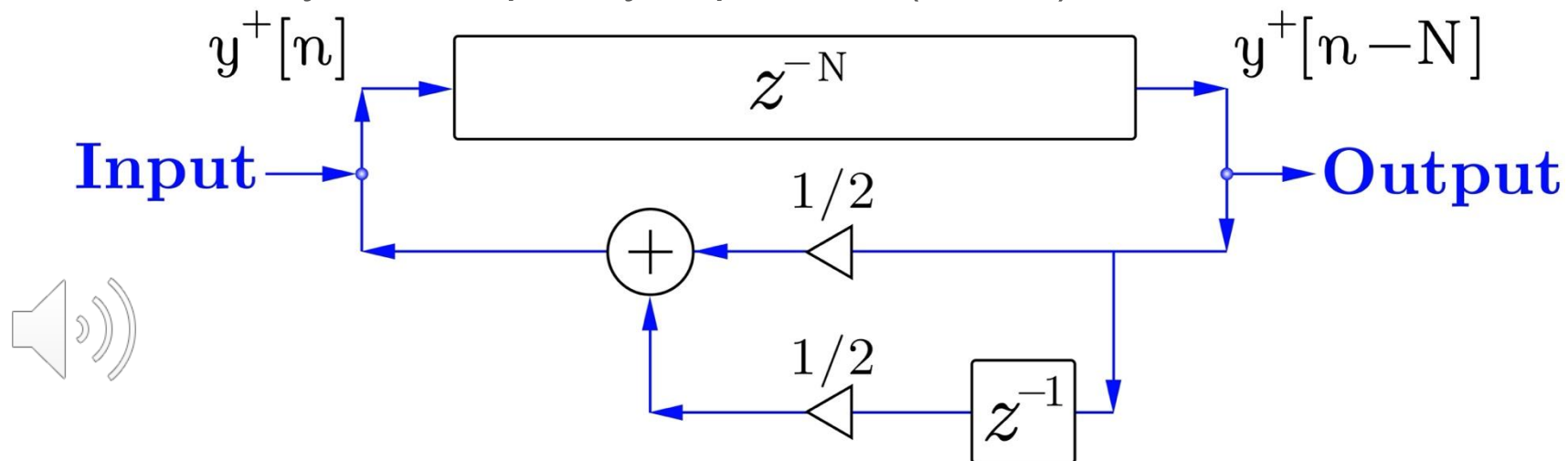
Karplus, K. and Strong, A., 1983. Digital synthesis of plucked-string and drum timbres. *Computer Music Journal*, 7(2), pp.43-55.

Karplus-Strong Algorithm

- As samples pass through the delay line, each sample is replaced by the average of itself and the previous sample, forming an attenuation filter in the feedback path.

$$H(z) = \frac{1}{1 - z^{-N}(1 + z^{-1})}$$

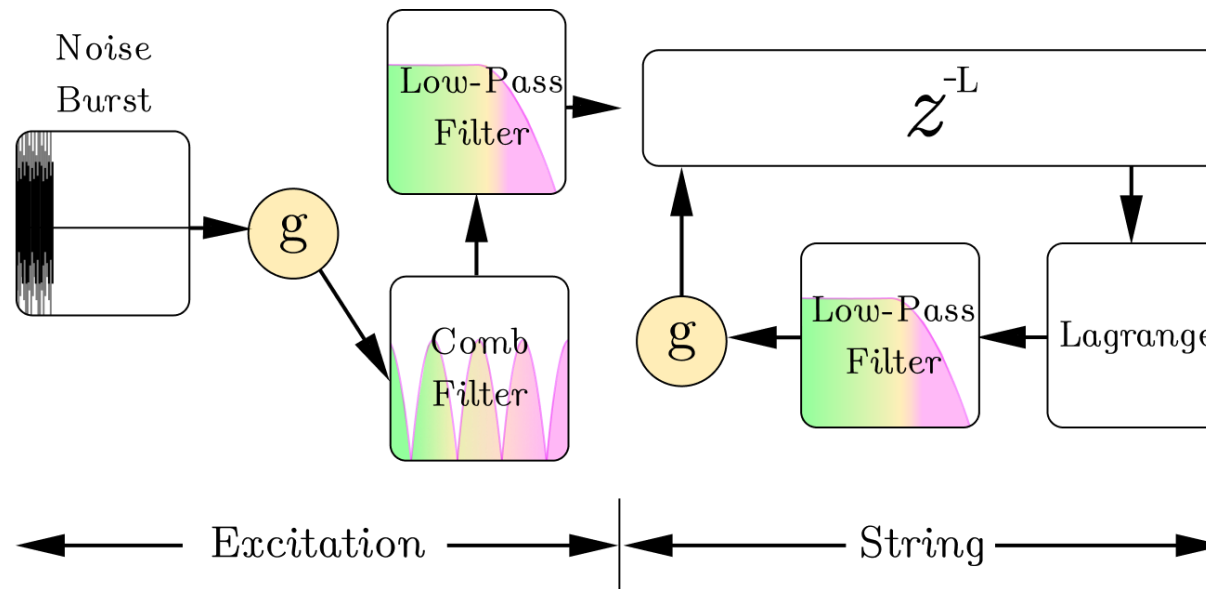
- It describes the decay and frequency dependant (timbre) characteristics of the string.



Karplus, K. and Strong, A., 1983. Digital synthesis of plucked-string and drum timbres. *Computer Music Journal*, 7(2), pp.43-55.

Karplus-Strong Algorithm (Extensions)

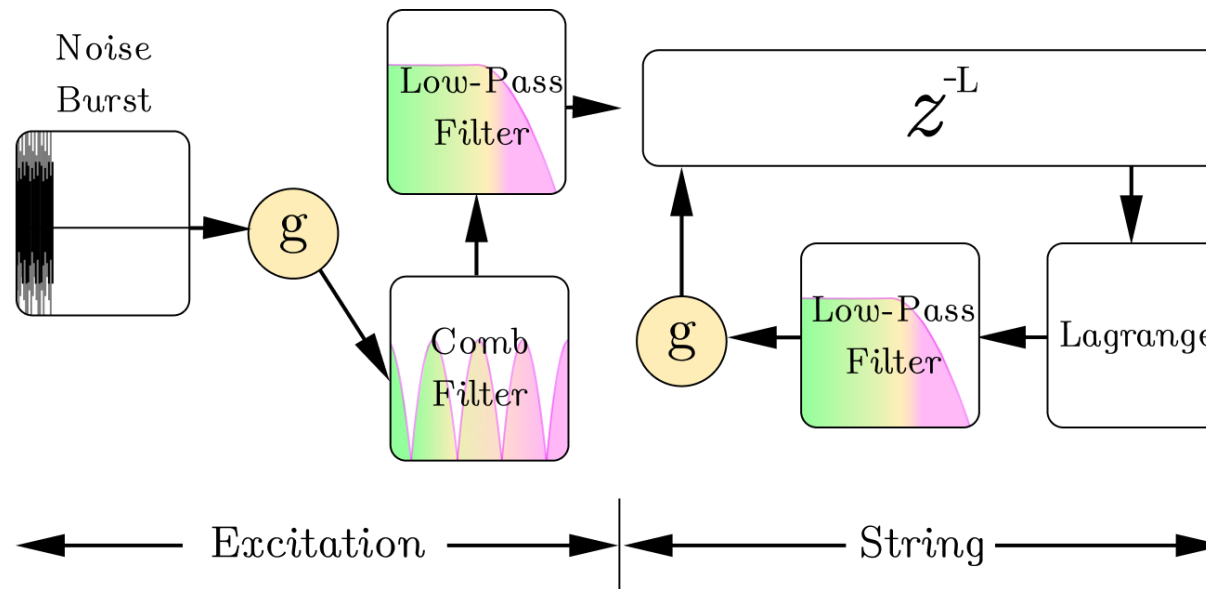
- Jaffe & Smith proposed filtering the input noise with:
 - a comb filter to simulate pluck position,
 - and a low-pass filter and gain stage to simulate playing dynamics.



Jaffe, D.A. and Smith, J.O., 1983. Extensions of the Karplus-Strong plucked-string algorithm. *Computer Music Journal*, 7(2), pp.56-69.

Karplus-Strong Algorithm (Extensions)

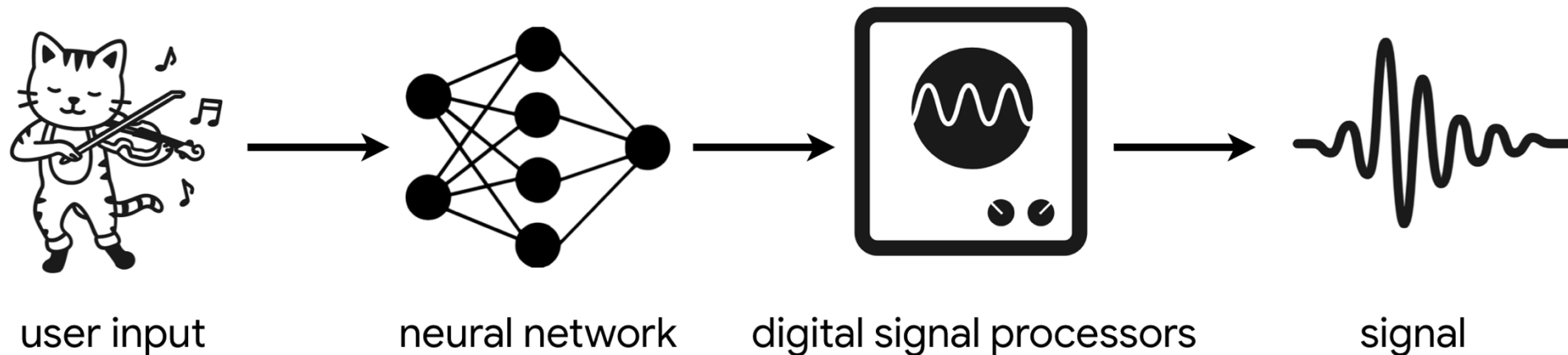
- Välimäki, Huopaniemi and Karjalainen proposed:
 - using a Lagrange interpolator (flat frequency response) for fractional delay lengths,
 - using a normalized first-order IIR, with a gain stage for decay control.



Välimäki, V. et al. 1996. Physical modeling of plucked string instruments with application to real-time sound synthesis. JAES.

DDSP & Timbre-Transfer

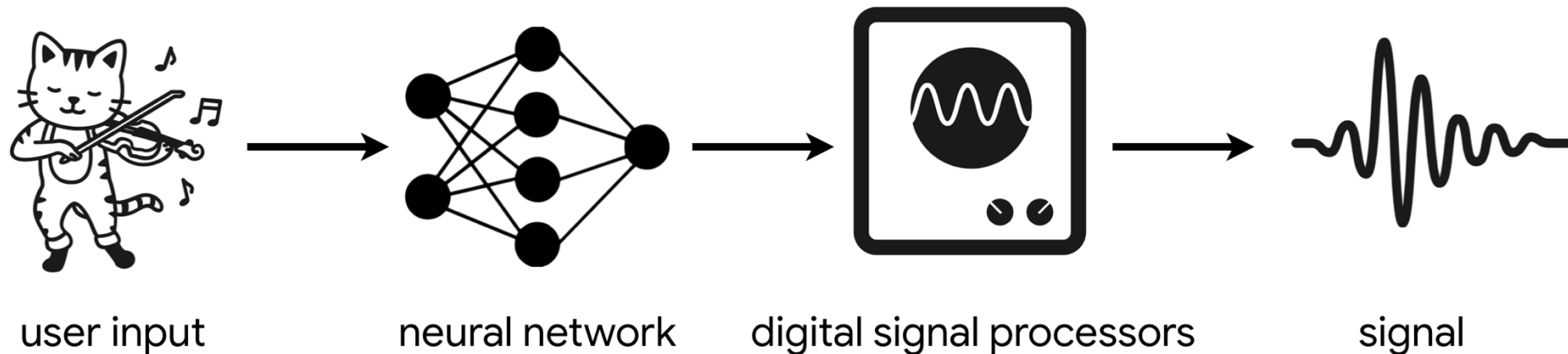
- Engel et al., proposed backpropagating through the parameters of Digital Signal Processors (hence, DDSP, differentiable DSP).
- Allows Self-supervised learning for the synthesiser parameter inversion problem.
- Since the output is limited to the DSP, training time is reduced substantially.



Engel, J., Gu, C. and Roberts, A., 2020. DDSP: Differentiable Digital Signal Processing. In *ICLR*.

DDSP & Timbre-Transfer

- In the original implementation, a neural network learnt the amplitude of harmonics and noise filterbanks from a spectral representation (timbre).
- While Loudness and f_0 are extracted separately.
- In result, a user-input can be transferred to the timbre of a previously learnt instrument.



Engel, J., Gu, C. and Roberts, A., 2020. DDSP: Differentiable Digital Signal Processing. In *ICLR*.

DDSP & Timbre-Transfer



Source: <https://www.youtube.com/watch?v=bXBliLjImio>

DDSP & Timbre-Transfer

- The last five years have seen numerous applications to other DSP:
 - FM Synthesis
 - Waveshaping
 - Physical Modelling
- Some general-purpose libraries:
 - FLAMO, a Frequency-sampling library for LTI systems
 - PhilTorch, library for Time-Domain filters (LTI and LTV)

Hayes, B., Saitis, C. and Fazekas, G., 2021. Neural Waveshaping synthesis. In *ISMIR*.

Caspe, F., McPherson, A. and Sandler, M., 2022, December. DDX7: Differentiable FM Synthesis of Musical Instrument Sounds. In *ISMIR*.

Südholt, D., Cámara, M., Xu, Z. and Reiss, J.D., 2023. Vocal Tract Area Descent Estimation by Gradient Descent. In *DAFx23*.

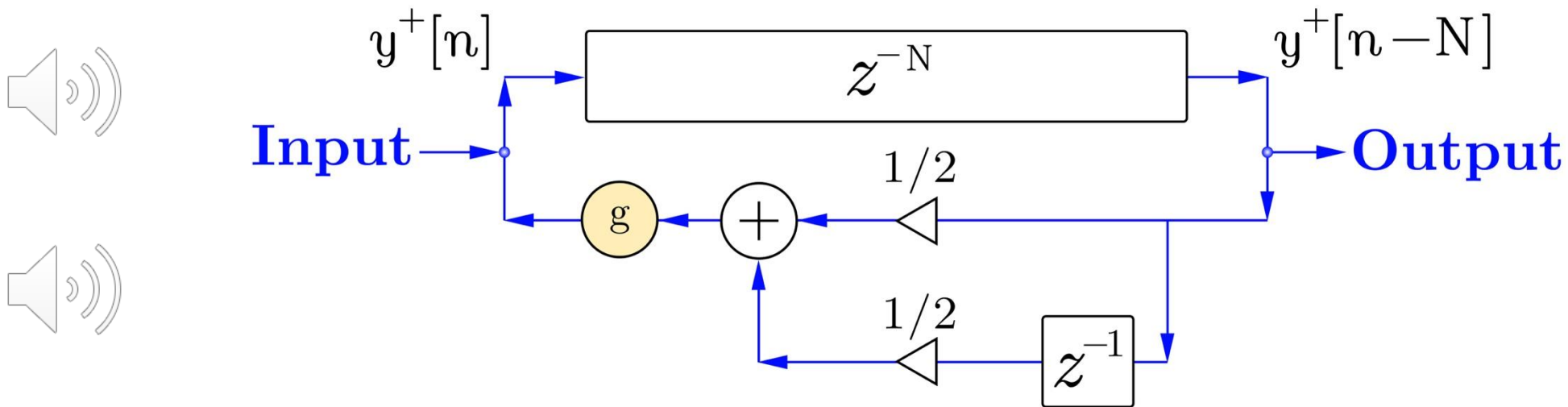
Mezza, A.I., Giampiccolo, R. and Bernardini, A., 2025. Differentiable Scattering Delay Networks for Artificial Reverberation. In *DAFx25*.

Dal Santo, G., et al., 2025, April. FLAMO: An Open-Source Library for Frequency-Domain Differentiable Audio Processing. In *ICASSP*.

Yu, C.Y. and Fazekas, G., 2025. Accelerating Automatic Differentiation of Direct Form Digital Filters. *EurIPS*

Previous Work

- In the accompanying materials of Hayes et al.
 - LTI implementation with frequency sampling,
 - Where *decay*, as a gain stage in the loop, can be optimised.



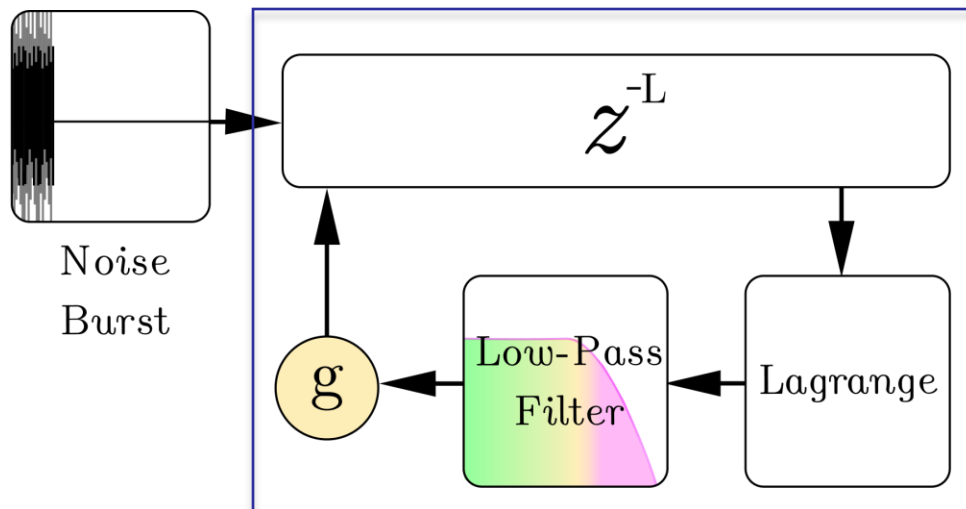
Source: https://intro2ddsp.github.io/physical-modeling/wave_equation.html

Differentiable Karplus-Strong – why?

- We propose using Karplus-Strong for timbre-transfer because:
 - It is very efficient;
 - It can simulate strings with a fraction of the DSP.
 - It offers a strong structural prior.
 - Hence, we hypothesize:
 - the neural network will need fewer parameters,
 - train faster,
 - and transients will be better represented.

Differentiable Karplus-Strong – how?

- IIR systems have dense computational graphs if implemented naively, but we can do:
 - Frequency Sampling methods.
 - Time-Domain methods, handle time-varying systems natively
 - but do not (currently) allow for additional filtering in feedback paths.



We can fold the
recursion

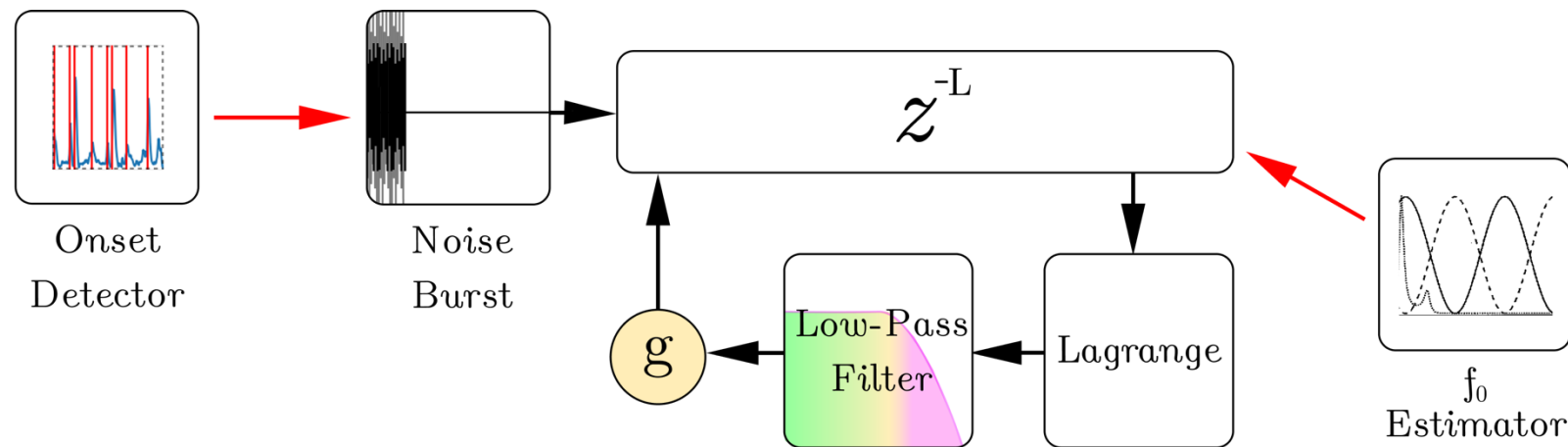
into one big
IIR!

$$H_{ks}(z) = \frac{z^{-L}}{1 - g(H_{IIR}(z) * H_{Lagrange}(z) * z^{-L})}$$

Yu, C.Y., Mitcheltree, C., et al., 2024. Differentiable All-Pole Filters for Time-Varying Audio Systems. In *DAFx24*

Differentiable Karplus-Strong – how?

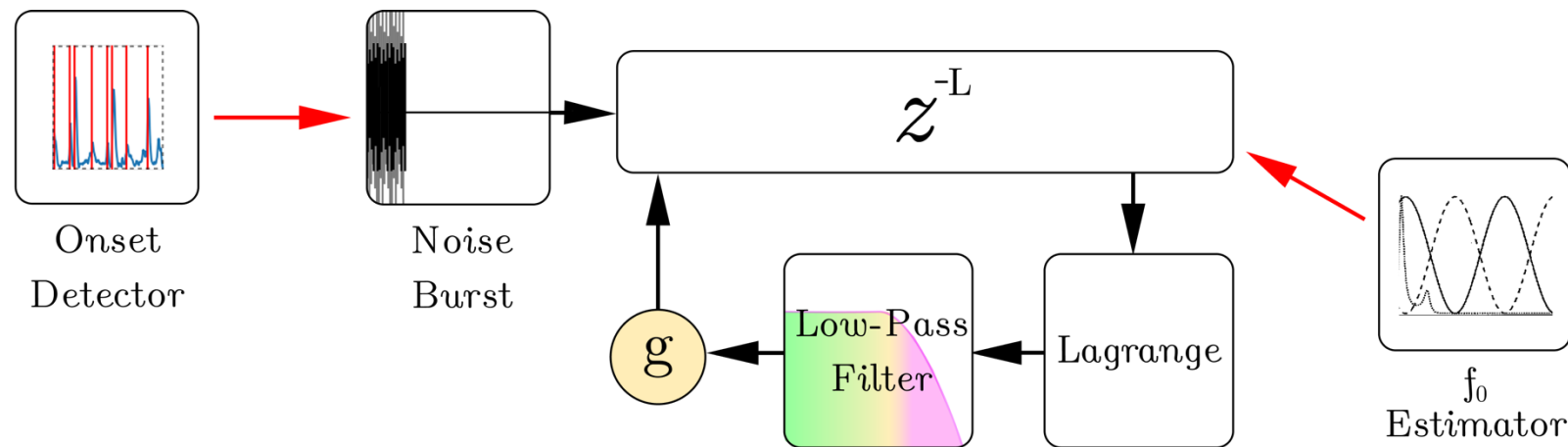
- Lastly, we need:
 - an onset detector, to place noise bursts.
 - an f_0 estimator, to define delay line length L .



McFee, B. et al. 2023. librosa/librosa: 0.10.0. *zenodo*.

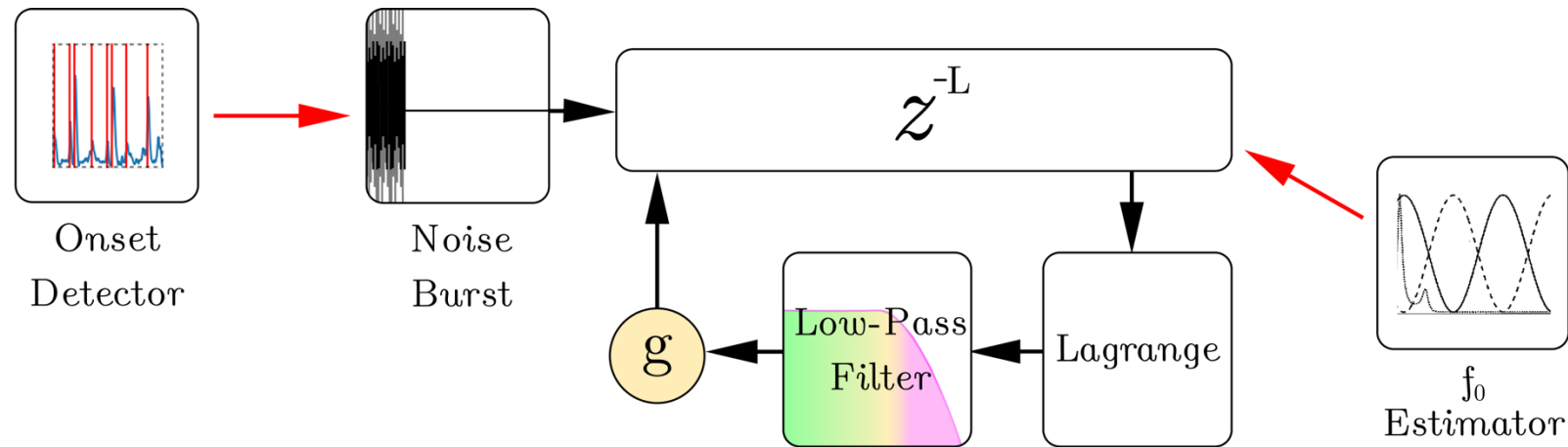
Differentiable Karplus-Strong – how?

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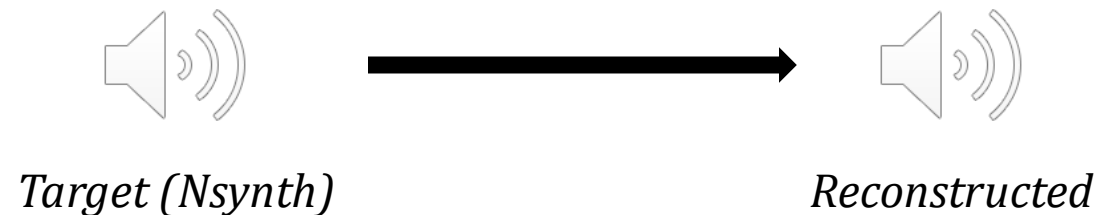


McFee, B. et al. 2023. librosa/librosa: 0.10.0. *zenodo*.

Differentiable Karplus-Strong – how?

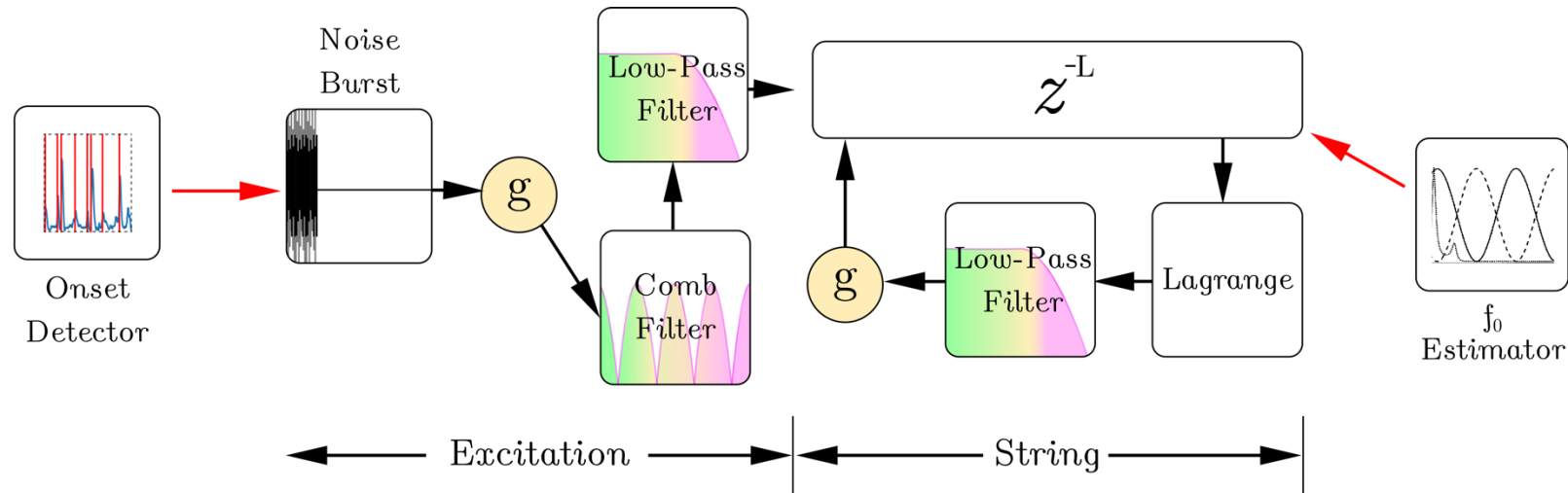


* Only the coloured blocks are optimized.

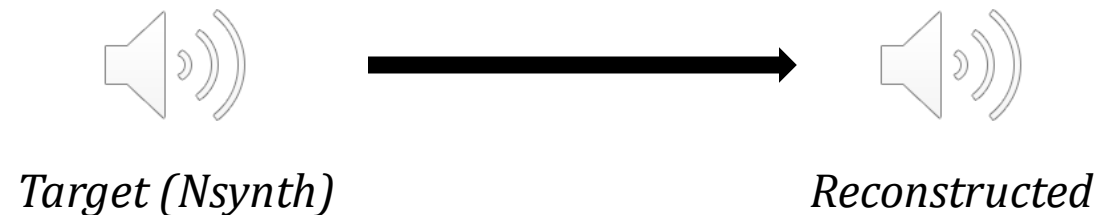


Engel, J. et al. 2017. Neural audio synthesis of musical notes with wavenet autoencoders. In *ICML*

Differentiable Karplus-Strong – how?



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Engel, J. et al. 2017. Neural audio synthesis of musical notes with wavenet autoencoders. In *ICML*

Experiment 1.

- Reconstruction task on acoustic Guitar subset of Nsynth ($N = 240$).
- Gradient Descent vs. Genetic Algorithm
 - We limit the Genetic Algorithm to *approx.* 10x more computationally expensive.

Table 1: Baseline Comparison. Lower values, in bold, are better.

Baseline	$\mathcal{L}_{\text{MSS}}$	MFCC	Encodec	MS-CLAP
Gradient Descent	0.79 ± 0.11	2.32 ± 1.50	19.89	111.30
Genetic Algorithm	1.43 ± 0.31	4.10 ± 1.75	216.93	346.93

Experiment 2.

Table 2: $\mathcal{L}_{\text{MSS}}$ mean difference to overall mean (0.62) across MIDI velocities and octaves for our GD baseline. Negative values are better.

↓ Oct. Vel. →	25	50	75	100	127	mean
E2–E3	+0.27	+0.23	+0.33	+0.30	+0.36	+0.30
E3–E4	-0.02	+0.07	-0.02	+0.07	+0.01	+0.02
E4–E5	-0.18	-0.10	-0.05	-0.13	-0.08	-0.11
E5–E6	-0.19	-0.26	-0.27	-0.18	-0.11	-0.20
mean	-0.03	-0.01	+0.00	+0.02	+0.05	0.00

- Karplus-Strong performs inconsistently across dynamics and pitches.
 - Worse at lower octaves
 - Worse at higher dynamics

Discussion

In acoustic guitars:

- Inharmonicities and non-linearities are more prominent in the low end and at higher dynamics
- Helmholtz and plate modes (guitar body) peak at 100 and 200 Hz.

But our Karplus-Strong Algorithm does not model any of these.

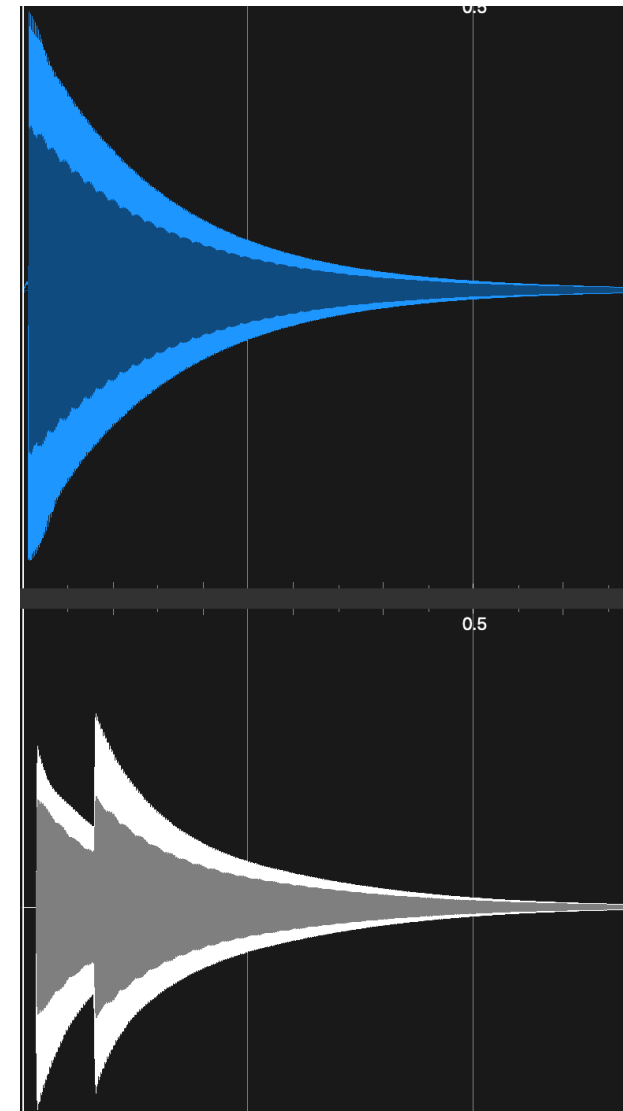


V. Välimäki and T. Tolonen, 1998. "Development and Calibration of a Guitar Synthesizer," in *JAES*.
J. S. Abel et al., 2010 "Robust, efficient design of allpass filters for dispersive string sound synthesis," in *Signal Proc. Lett.*

Discussion

We are also limited by our choice of:

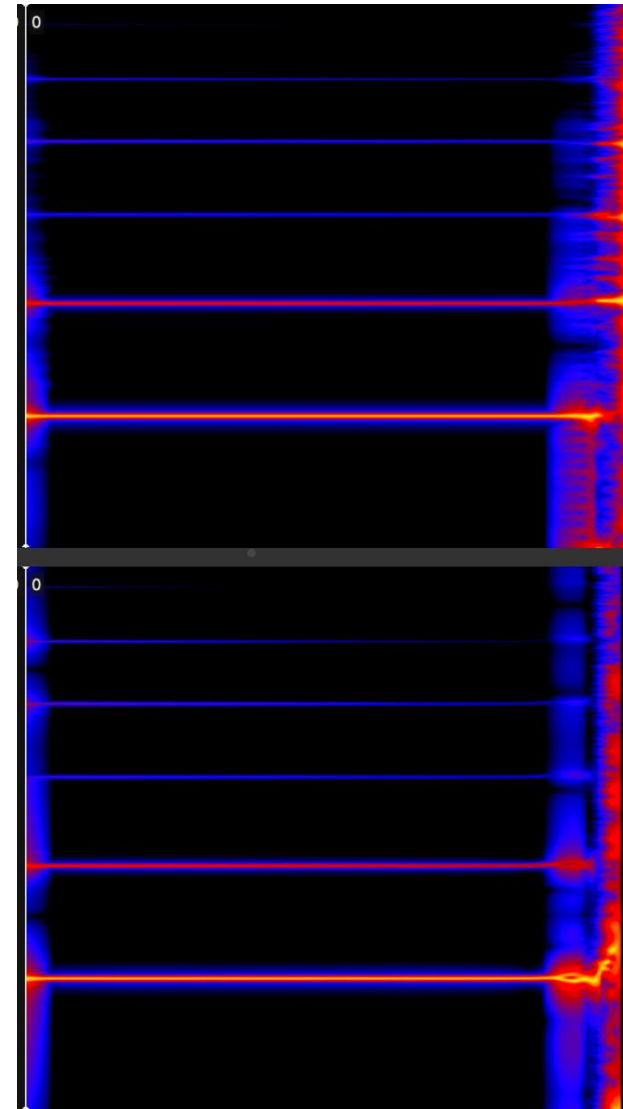
- onset detector;
 - It might double trigger,
 - or it might trigger late.



Discussion

We are also limited by our choice of:

- onset detector;
 - It might double trigger,
 - or it might trigger late.
- f_0 predictor;
 - a change of resonance might affect the pitch on spectrum.



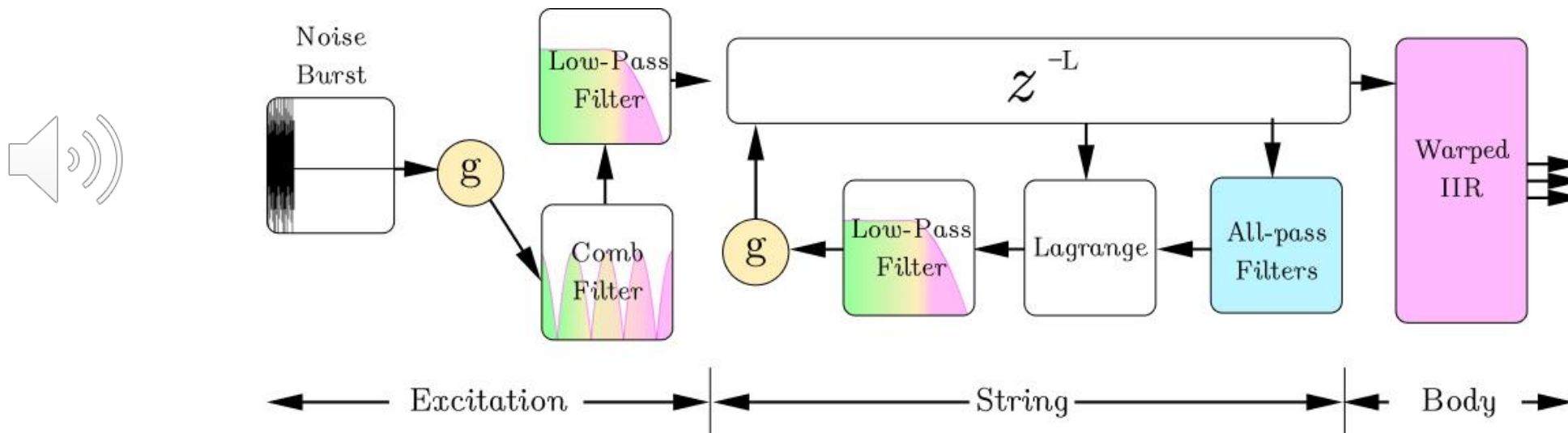
Future Work

Should address two orthogonal sets of issues:

- Reconstruction capacity of Karplus-Strong.
- Embedding excitation and f_0 in the neural network.

Future Work: Reconstruction Capacity

- Parallel sections of warped IIRs have been proposed to model instrument bodies.
- Allpass filters in the loop can efficiently model inharmonicity



B. Bank, 2007 "Direct design of parallel second-order filters for instrument body modeling," in *ICMC*.

J. S. Abel et al., 2010 "Robust, efficient design of allpass filters for dispersive string sound synthesis," in *Signal Proc. Lett.*

Future Work: Embedding excitation and f_0

- DDSP can use parameter losses (MAE on parameter sets) on synthetic data.
- The neural network could firstly be trained on parameter losses to obtain:
 - Excitation times,
 - f_0 .
- Alternatively, we could fine-tune neural networks for onset and f_0
- One problem arises, our delay line length is not differentiable (gray-box).

Engel, J. et al., 2020. Self-supervised pitch detection by inverse audio synthesis. In *ICML*

Masuda, N. and Saito, D., 2023. Improving semi-supervised differentiable synthesizer sound matching. In *TASLP*

Han, H. Lostanlen, V., Lagrange, M., 2024 Learning to solve inverse problems for perceptual sound matching. In *TASLP*

Contributions

- A time-domain time-variant differentiable implementation for an extended Karplus-Strong Algorithm.
- We found that extensions are needed to consistently model the full dynamic and pitch range of the guitar.
- We observed onset and fundamental frequency predictors can limit timbre-transfer tasks.

Questions?



Audio examples,
presentation, paper, tutorials
& code



Let's connect on
LinkedIn!